DSP · L04

LCCDE and convolution

Cheat sheet · Day 04 · Tue 27 Oct · pack D2

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

```
LCCDE  sum a_p y(n-p) = sum b_q x(n-q), a0=1, order k
MODES  lambda^k + a1 lambda^(k-1)+...+ak = 0   (y-coeffs ONLY; no y-terms => no modes
  distinct: C1 l1^n + C2 l2^n | double: C1 l^n + C2 n l^n (l=1: C1 + C2 n)
  complex r e^{+-j th}: r^n (A cos th n + B sin th n)
PARTICULAR  same form as x: K a^n (K n a^n if a is a root), K, K1 cos+K2 sin; sub int
TOTAL  y = yh + yp; y(0..k-1) BY RECURSION from ICs; solve for C's; attach u(n); chec
RECURSION  y(n) = -a1 y(n-1) - ... + b0 x(n) + b1 x(n-1)...  (flip y signs)
h(n)  x=delta, zero ICs. FIR: h=b_n. IIR: modes + h(0..k-1) by recursion.
      m>=k: extra delta(n-i) terms, i=0..m-k
CONV  y(n)=sum_k h(k)x(n-k); steps -> limits on k AND range of n
  finite: start = nx+nh, length Nx+Nh-1, check sum y = (sum x)(sum h)
  sum_{k1}^{k2} r^k = (r^k1 - r^(k2+1))/(1-r)
  a^n u * b^n u = (a^(n+1)-b^(n+1))/(a-b) u(n);  a^n u * a^n u = (n+1) a^n u(n)
  check y(0)=h0x0, y(1)=h0x1+h1x0
PROPERTIES  causal: h=0 n<0 | stable: sum|h|<inf (all |lambda|<1) | FIR always stable
  |lambda|=1 or u(n),(n+1)u(n),cos u(n) => UNSTABLE | static: h=K delta(n)
INTERCONNECT  cascade h1*h2 (any order); parallel h1+h2; x*delta(n-d)=x(n-d)
```